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Inventor(s)

HSU; Ming-Chieh et al.

CHEMICAL DISPENSING SYSTEM

Abstract

A chemical dispensing system is capable of simultaneously supplying a semiconductor processing chemical for production and testing through the use of independent chemical supply lines, which reduces production downtime of an associated semiconductor process, increases throughput and capability of the semiconductor process, and/or the like. Moreover, the capability to simultaneously supply the semiconductor processing chemical for production and testing allows for an increased quantity of semiconductor processing chemical batches to be tested with minimal impact to production, which increases quality control over the semiconductor processing chemical. In addition, the independent chemical supply lines may be used to supply the semiconductor processing chemical to production while independently filtering semiconductor processing chemical directly from a storage drum through a filtration loop.

Inventors: HSU; Ming-Chieh (Hsinchu City, TW), CHEN; Yung-Long (New Taipei City, TW), CHIANG; Fang-Pin (Taichung City, TW), YANG; Feng-An (Hsinchu City, TW), HSU; Ching-Jung (Hsinchu City, TW), LAI; Chi-Tung (Hsinchu, TW)

Applicant: Taiwan Semiconductor Manufacturing Company, Ltd. (Hsinchu, TW)

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Background/Summary

RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 18/447,353, filed Aug. 10, 2023, which is a continuation of U.S. patent application Ser. No. 16/887,647, filed May 29, 2020 (now U.S. Pat. No. 11,817,329), the contents of which are incorporated herein by reference in their entireties.

BACKGROUND

[0002] Semiconductor processing chemicals may be supplied via a chemical dispensing system for various types of semiconductor processes, such as wet etching, lithography, chemical vapor deposition, physical vapor deposition, and/or the like. A chemical dispensing system may include various types of storage containers for storing semiconductor processing chemicals. For example, a chemical dispensing system may include a storage drum for storing bulk semiconductor processing chemical, a day tank for storing semiconductor processing chemical for daily use, and/or the like. The semiconductor processing chemical may be pumped from the day tank to a production point of use, where the semiconductor processing chemical is used as part of a semiconductor process.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIGS. 1A and 1B are diagrams of an example chemical dispensing system described herein.

[0005] FIGS. 2A-2R are diagrams of example implementations described herein.

[0006] FIGS. 3A and 3B are diagrams of one or more example implementations described herein.

[0007] FIG. 4 is a diagram of example components of one or more devices of FIGS. 1A, 1B, 2A-2R, 3A, and 3B.

[0008] FIG. 5 is a flowchart of an example process for dispensing a semiconductor processing chemical.

[0009] FIG. 6 is a flowchart of an example process for filtering a semiconductor processing chemical.

[0010] FIG. 7 is a flowchart of an example process for dispensing a semiconductor processing chemical.

DETAILED DESCRIPTION

[0011] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature

over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0012] Wafer defects can occur as a result of using a semiconductor processing chemical having impurities and other defects in a semiconductor process. While a chemical dispensing system may include one or more filters to filter out contaminants and other impurities, not every batch of semiconductor processing chemical may be inspected because the chemical dispensing system might not be capable of supporting testing and production in parallel. To test a batch of semiconductor processing chemical, supply to a production process is stopped, and the chemical dispensing system is transitioned to supply a semiconductor processing test. The process of switching the chemical dispensing system from production to test and back, and the process of testing itself, are time-consuming processes that would greatly decrease the productivity of the semiconductor processes supplied by the chemical dispensing system if each new batch of semiconductor processing chemical were tested.

[0013] Some implementations described herein provide a chemical dispensing system that is capable of simultaneously supplying a semiconductor processing chemical for production and testing. The chemical dispensing system includes a first pump that pumps a semiconductor processing chemical from a day tank and through a first chemical supply line, and a second pump that pumps a semiconductor processing chemical from a storage drum and through a second chemical supply line. The first chemical supply line and the second chemical supply line are independent chemical supply lines that are capable of being configured to simultaneously supply the semiconductor processing chemical to a production point of use (POU) for a semiconductor process and a test POU for testing.

[0014] In this way, the independent chemical supply lines permit production and testing to be performed in parallel, which reduces production downtime of the semiconductor process, increases throughput and capability of the semiconductor process, and/or the like. Moreover, the capability to simultaneously supply the semiconductor processing chemical for production and testing allows for an increased quantity of semiconductor processing chemical batches to be tested with minimal impact to production, which increases quality control over the semiconductor processing chemical.

[0015] In addition, the independent chemical supply lines may be used to supply the semiconductor processing chemical to production while independently filtering semiconductor processing chemical directly from a storage drum instead of from a day tank. Since the supply of the semiconductor processing chemical in the day tank is consumed for daily production, it is difficult to maintain an even supply quality of the semiconductor processing chemical due to uneven filtering. Semiconductor processing chemical that is consumed early on may be filtered fewer times relative to semiconductor processing chemical that is consumed later on, which may lead to inconsistent levels of quality of the semiconductor processing chemical. With the independent chemical supply lines, the chemical dispensing system is capable of supplying semiconductor processing chemical from the storage drum to a filtration loop prior to being provided to the day tank. In this way, the semiconductor processing chemical in the day tank is filtered evenly, which increases the quality of the semiconductor processing chemical.

[0016] FIGS. 1A and 1B are diagrams of an example chemical dispensing system **100** described herein. In some implementations, chemical dispensing system **100** may also be referred to as a chemical dispense unit (CDU), a chemical dispense system (CDS), a chemical supply system, or another system that is capable of supplying a semiconductor processing chemical for one or more types of semiconductor processes, such as wet etching, lithography, chemical vapor deposition,

physical vapor deposition, and/or the like. The semiconductor processing chemical may include a liquid, a gas, a slurry, and/or another form of chemical. Examples of semiconductor processing chemicals that may be supplied by chemical dispensing system **100** include (but are not limited to) argon, oxygen, hydrofluoric acid, phosphoric acid, ammonium fluoride, various halides, various hydrides, various nitrides, precursor materials, buffering agents, combinations thereof, and/or many other chemicals used in various semiconductor processes.

[0017] As shown in FIG. **1A**, chemical dispensing system **100** may include various components, such as one or more storage drums **102**, one or more day tanks **104**, a plurality of valves **106**, a plurality of pumps **108**, one or more production point of use (POU) manifolds **110**, one or more chemical test POU manifolds **112**, a plurality of filters **114**, one or more pressure tanks **116**, one or more controllers **118**, and/or the like. Moreover, chemical dispensing system **100** may include various types and configurations of plumbing fixtures for transporting, supplying, and/or otherwise providing a semiconductor processing chemical to the various components of chemical dispensing system **100**, such as pipes (e.g., copper pipes, galvanized steel pipes, cast iron pipes, polyvinyl chloride (PVC) pipes, and/or the like), fittings (e.g., couplings, elbows, reducers, expanders, unions, cross fittings, tee fittings, Y fittings, and/or the like), and/or the like.

[0018] Storage drums **102** (e.g., storage drum **102a**, storage drum **102b**, and/or the like) include one or more storage drums, one or more storage containers, one or more storage vessels, or one or more other types of apparatuses capable of storing semiconductor processing chemical (e.g., bulk or raw semiconductor processing chemical). In some implementations, storage drums **102** may be capable of storing semiconductor processing chemical for various duration lengths, such as a plurality of days, a plurality of weeks, a plurality of months, and/or the like. Storage drums **102** may be capable of holding various amounts of semiconductor processing chemical, such as **100** liters, **200** liters, and/or the like.

[0019] In some implementations, a storage drums **102** may be capable of operating in various modes, such as a supply mode in which the storage drum **102** is supplying semiconductor processing chemical to various components of chemical dispensing system **100**, a standby mode in which the storage drum **102** is not supplying semiconductor processing chemical, and/or the like. In some implementations, a storage drum **102** (e.g., storage drum **102a**) may be configured to operate in a supply mode while another storage drum **102** (e.g., storage drum **102b**) is configured to operate in a standby mode. In this way, a storage drum **102** in supply mode may supply semiconductor processing chemical to various components of chemical dispensing system **100** while another storage drum **102** in standby mode may be refilled with semiconductor processing chemical, may be filtered (e.g., the semiconductor processing chemical in the storage drum **102** may be circulated through a filtration loop), and/or the like.

[0020] Day tank **104** includes a tank, a drum, a storage container, a storage vessel, or another type of apparatus capable of storing a semiconductor processing chemical for daily use at the one or more production POU manifolds **110**, the chemical test POU manifold **112**, and/or at one or more other components. In some implementations, day tank **104** may be sized to hold an amount of semiconductor processing chemical that is expected to be consumed for daily use, such as **500** liters or more. In some implementations, day tank **104** has a capacity equal to or greater than a capacity of a storage drum **102**.

[0021] In some implementations, semiconductor processing chemical is provided to day tank **104** from one or more storage drums **102**. In some implementations, the semiconductor processing chemical is filtered through one or more filters **114** prior being added to day tank **104**. In some implementations, the semiconductor processing chemical in day tank **104** may be diluted (e.g., relative to the bulk or raw semiconductor processing chemical stored in storage drums **102**), may be mixed with additives, and/or the like prior to use at the one or more production POU manifolds **110**, the chemical test POU manifold **112**, and/or the like.

[0022] Valves **106** include various types of plumbing valves made from various types of materials

such as copper plumbing valves, PVC valves, cast iron valves, and/or the like. A valve **106** may be capable of permitting the passage of semiconductor processing chemical (e.g., when the valve **106** is configured in an open position) and restricting or blocking the passage of semiconductor processing chemical (e.g., when the valve **106** is configured in a closed position). In this way, valves **106** may be used to direct the flow of semiconductor processing chemical along various flow paths in chemical dispensing system **100**, may be used to block or restrict the flow of semiconductor processing chemical so that components such as pumps **108** or filters **114** may be bypassed, removed for maintenance, or replaced, and/or the like. Examples of valves **106** include ball valves, butterfly valves, check valves, pneumatic valves, diaphragm valves, and/or the like.

[0023] Pumps **108** include various types of pumps capable of pumping, supplying, and/or otherwise causing semiconductor processing chemical to flow through chemical dispensing system **100**. For example, pumps **108** may include centrifugal pumps, in-line pumps, reciprocating pumps, rotary pumps, and/or other types of pumps. Chemical dispensing system **100** may include a plurality of pumps **108** to supply or provide semiconductor processing chemical through a plurality of independent chemical supply lines. In this way, chemical dispensing system **100** is capable of simultaneously supplying or providing semiconductor processing chemical from a plurality of different sources (e.g., day tank **104** and a storage drum **102**) to a plurality of different destinations (e.g., a production POU manifold **110** and a chemical test POU manifold **112**).

[0024] In some implementations, pump **108a** and pump **108b** may be configured to pump, supply, and/or otherwise cause semiconductor processing chemical in day tank **104** to flow through a chemical supply line of chemical dispensing system **100**. In some implementations, pump **108a** may be a primary pump for day tank **104**, and pump **108b** may be a secondary or backup pump for day tank **104** that is used if pump **108a** fails, experiences abnormal behavior, or is in the process of being replaced or serviced. Moreover, pump **108c** may be configured to pump, supply, and/or otherwise cause semiconductor processing chemical in storage drums **102** to flow through a chemical supply line and/or a filtration loop of chemical dispensing system **100** to various locations, such as day tank **104**, production POU manifold **110**, chemical test POU manifold **112**, and/or the like. Pump **108c** is different from and may operate (or be operated) independent of pump **108a** and pump **108b**.

[0025] Production POU manifold(s) **110** includes various types of manifolds or manifold systems capable of supplying or providing a semiconductor processing chemical to a plurality of semiconductor processing POUs for production use. Examples of semiconductor processing POUs include (but are not limited to) a chemical vapor deposition processing chamber, a physical vapor deposition processing chamber, a spin coating machine, a wet etch bath, and/or another location in a semiconductor processing facility where the semiconductor processing chemical is used in a semiconductor process. Chemical test POU manifold **112** includes various types of manifolds or manifold systems capable of supplying or providing a semiconductor processing chemical to a plurality of testing POUs. Examples of testing POUs include (but are not limited to) a test fixture, a test chamber, a lab, and/or the like.

[0026] Filters **114** include various types of filters capable of filtering contaminants, foreign material, and/or other types of impurities from a semiconductor processing chemical stored and/or supplied by chemical dispensing system **100**. For example, filters **114** may include membrane filters, pleated filters, cylindrical filters, mesh filters, and/or other types of filters capable of filtering liquids and/or gasses. In some implementations, subsets of the filters **114** included in chemical dispensing system **100** may be configured to perform various types of filtrations. For example, one or more filters **114** may be configured to perform loop filtration, recirculation filtration, or another type of filtration in which semiconductor processing chemical is circulated through a filtration loop. As another example, one or more filters **114** may be configured to perform final stage filtration, a one-pass filtration, POU filtration, or another type of filtration in which semiconductor processing chemical is filtered prior to being supplied to a production POU

manifold **110** for use in a semiconductor process or a chemical test POU manifold **112** for testing. In some implementations, a portion of the semiconductor processing chemical processed through a final stage filtration may be returned to day tank **104** to ensure an even flow of semiconductor processing chemical to production POU manifold(s) **110** and/or chemical test POU manifold **112**. [0027] Pressure tank(s) **116** include one or more tanks, vessels, containers, or other types of apparatuses capable of holding semiconductor processing chemical while the semiconductor processing chemical is pressurized using a pressurized gas (e.g., nitrogen, oxygen, and/or the like). For example pressure tank(s) **116** may each hold 20 liters of semiconductor processing chemical or another volume. The semiconductor processing chemical in pressure tank(s) **116** may be pressurized to maintain an even flow and supply of semiconductor processing chemical to production POU manifold(s) **110** and/or chemical test POU manifold **112**.

[0028] As shown in FIG. **1B**, chemical test POU manifold **112** may include a plurality of valves **106**, a chemical test supply line **120**, a production supply line **122**, a sampling line **124**, a flush drain line **126**, and/or one or more other components and combinations thereof. Chemical test supply line **120** may supply semiconductor processing chemical (e.g., a raw or bulk supply semiconductor processing chemical) to one or more testing POUs **128** via one or more valves **106**. Similarly, production supply line **122** may supply semiconductor processing chemical (e.g., a production semiconductor processing chemical) to one or more testing POUs **128** via one or more valves **106**. Valves **106** (e.g., POU valves) may be used to restrict flow of semiconductor processing chemical to particular testing POUs **128**, may be used to permit flow of semiconductor processing chemical to particular testing POUs **128**, may be used to restrict or permit flow to chemical test supply line **120**, production supply line **122**, and/or sampling line **124**, and/or the like. [0029] Sampling line **124** may be used to directly test the semiconductor processing chemical from production POU manifold(s) **110** and/or chemical test POU manifold **112** for increased quality control sampling. Flush drain line **126** may be used to drain or flush semiconductor processing chemical from chemical test POU manifold **112**. Flush drain line **126** may reduce and/or minimize the risk of dead zone residue contamination in chemical test POU manifold **112**.

[0030] The number and arrangement of components shown in FIGS. **1A** and **1B** are provided as an example. In practice, chemical dispensing system **100** may include additional components, fewer components, different components, or differently arranged components than those shown in FIGS. **1A** and/or **1B**. Additionally, or alternatively, a set of components (e.g., one or more components) of chemical dispensing system **100** may perform one or more functions described as being performed by another set of components of chemical dispensing system **100**.

[0031] FIGS. **2A-2R** are diagrams of example implementations **200** described herein. As shown in FIGS. **2A-2R**, example implementations **200** may include a chemical dispensing system, such as chemical dispensing system **100** illustrated and described above in connection with FIGS. **1A** and **1B**.

[0032] As shown in FIG. **2A**, chemical dispensing system **100** may operate in a production supply configuration, in which one or more components of chemical dispensing system **100** form a chemical supply line **202** that is configured to provide a semiconductor processing chemical from day tank **104** at least to production POU manifold(s) **110**. Chemical supply line **202** may be composed of various components included in chemical dispensing system **100**, such as one or more valves **106**, one or more filters **114**, one or more pressure tanks **116**, one or more pipes, one or more fittings, and/or one or more other types of plumbing fixtures. In the production supply configuration, one or more valves **106**, one or more pumps **108** (e.g., pump **108a**, pump **108b**, and/or the like), one or more filters **114**, one or more pressure tanks **116**, one or more pipes, one or more fittings, and/or one or more other plumbing fixtures included in chemical dispensing system **100** may be opened, closed, connected, and/or otherwise configured such that the semiconductor processing chemical in day tank **104** is directed along a flow path of chemical supply line **202** to production POU manifold(s) **110** and, in some cases, chemical test POU manifold **112**.

[0033] As shown in FIG. 2B, and by reference number **204**, controller **118** may determine that a test is to be performed. In some implementations, controller **118** may determine to perform a maintenance component test or another type of test that is performed to check the performance of chemical dispensing system **100** after a new component is added to chemical dispensing system **100**, after a component in chemical dispensing system **100** is replaced, after a chemical dispensing system **100** is repaired, and/or the like. In these cases, semiconductor processing chemical stored in day tank **104** and/or semiconductor processing chemical stored in a storage drum **102** may be provided to chemical test POU manifold **112** for sampling and testing as part of the test. In some aspects, controller **118** may determine to perform a semiconductor processing chemical test based on a new batch of semiconductor processing chemical being added to a storage drum **102**. In these cases, the semiconductor processing chemical in the storage drum **102** may be provided to chemical test POU manifold **112** for sampling and testing.

[0034] In some implementations, controller **118** may determine that a test is to be performed based on receiving an instruction to perform the test. In some implementations, the instruction may include an input from a user via an input device such as a mouse, a keyboard, a touch screen, a microphone to capture a voice command, and/or the like. In some implementations, the instruction may include an instruction received as input from another component included in chemical dispensing system **100** (e.g., another controller **118** and/or the like). In some implementations, the instruction may include an instruction received as input from a component that is remote from chemical dispensing system **100**, such as a remote computer, a cloud computing environment, and/or the like.

[0035] As further shown in FIG. 2B, and by reference number **206**, controller **118** may cause chemical dispensing system **100** to be transitioned from the production supply configuration to an independent production and test configuration. In some implementations, controller **118** causes chemical dispensing system **100** to be transitioned from the production supply configuration to an independent production and test configuration based on determining to perform a test (e.g., a test of chemical dispensing system **100**, a maintenance component test, a semiconductor processing chemical test, and/or the like). Controller **118** may cause chemical dispensing system **100** to be transitioned from the production supply configuration to an independent production and test configuration by automatically closing one or more valves **106**, opening one or more valves **106**, activating one or more pumps **108**, deactivating one or more pumps **108**, reconfiguring and/or rearranging one or more filters **114** (e.g., one or more filters **114** that provide a final stage filtration, a one-pass filtration stage, POU filtration, and/or the like), reconfiguring and/or rearranging one or more pipes, fittings, and/or other plumbing fixtures, and/or the like.

[0036] Semiconductor processing chemical may be supplied or provided to chemical test POU manifold **112** so that a test of the semiconductor processing chemical may be performed to generate wafer test data. The wafer test data may include information identifying one or more parameters for the semiconductor processing chemical. The one or more parameters may be indicators of the performance of the semiconductor processing chemical, may be indicators of the quality level of the semiconductor processing chemical, and/or the like. For example, the one or more parameters may indicate a purity of the semiconductor processing chemical, may indicate a level of impurities included in the semiconductor processing chemical, may indicate a coverage performance (e.g., a spin coating coverage performance), may indicate one or more measurements of a wafer (e.g., a feature size measurement, a structure measurement, and/or the like), may indicate an etch rate, and/or may indicate one or more other indicators of performance and/or quality of the semiconductor processing chemical. In some implementations, controller **118** may determine whether the one or more parameters for the semiconductor processing chemical satisfy one or more associated thresholds (e.g., a purity threshold, a level of impurities threshold, one or more performance thresholds, and/or the like) and may selectively cause another test to be performed based on a result of the determination. Moreover, if controller **118** determines that a parameter for

the semiconductor processing chemical does not satisfy a threshold, controller **118** may cause the semiconductor processing chemical to be filtered for one or more cycles prior to performing the other test.

[0037] FIGS. 2C-2R illustrate various example independent production and testing configurations for chemical dispensing system **100**. As shown in FIGS. 2C-2R, in an independent production and testing configuration, chemical dispensing system **100** may be configured such that chemical supply line **202** and another chemical supply line **208** are configured to simultaneously and independently provide and/or supply a semiconductor processing chemical to production POU manifold(s) **110** and chemical test POU manifold **112**. Chemical supply line **202** and another chemical supply line **208** may independently provide and/or supply a semiconductor processing chemical in that chemical supply line **202** and chemical supply line **208** supply the semiconductor processing chemical from independent sources and using different valves **106**, different pumps **108**, different filters **114**, and/or different pipes such that chemical supply line **202** and chemical supply line **208** supply the semiconductor processing chemical using different flow paths through chemical dispensing system **100**.

[0038] In particular, chemical supply line **202** is configured to provide and/or supply a semiconductor processing chemical from day tank **104**, and chemical supply line **208** is configured to provide and/or supply a semiconductor processing chemical from a storage drum **102** (e.g., storage drum **102a**, storage drum **102b**, and/or the like). Moreover, chemical supply line **202** is configured to provide and/or supply a semiconductor processing chemical from day tank **104** through the use of pump **108a** or pump **108b**, whereas chemical supply line **208** is configured to provide and/or supply a semiconductor processing chemical from a storage drum **102** through the use of pump **108c**. In these cases, pump **108a** or pump **108b** may pump and/or otherwise cause the semiconductor processing chemical in day tank **104** to traverse along the flow path of chemical supply line **202**, and pump **108c** may pump and/or otherwise cause the semiconductor processing chemical in the storage drum **102** to traverse along the flow path of chemical supply line **208**. In this way, chemical supply line **202** may supply the semiconductor processing chemical in day tank **104** in a manner in which the use of the semiconductor processing chemical in day tank **104** does not impact or affect the use and supply of the semiconductor processing chemical in the storage drum **102**. Similarly, chemical supply line **208** may supply the semiconductor processing chemical in the storage drum **102** in a manner in which the use of the semiconductor processing chemical in the storage drum **102** does not impact or affect the use and supply of the semiconductor processing chemical in day tank **104**.

[0039] As further shown in FIGS. 2C-2R, in an independent production and testing configuration, chemical supply line **202** and chemical supply line **208** may be configured to supply semiconductor processing chemical to different POU manifolds. For example, chemical supply line **202** may be configured to provide semiconductor processing chemical from day tank **104** to one or more production POU manifolds **110**, whereas chemical supply line **208** may be configured to provide semiconductor processing chemical from a storage drum **102** to chemical test POU manifold **112**. As another example, chemical supply line **202** may be configured to provide semiconductor processing chemical from day tank **104** to chemical test POU manifold **112**, whereas chemical supply line **208** may be configured to provide semiconductor processing chemical from a storage drum **102** to one or more production POU manifolds **110**.

[0040] As shown in FIG. 2C, an example independent production and testing configuration may include chemical supply line **202** being configured to provide a semiconductor processing chemical from day tank **104** to one or more production POU manifolds **110** through one or more valves **106**, pump **108a**, a pressure tank **116**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like. Moreover, in the example independent production and testing configuration illustrated in FIG. 2C, chemical supply line **208** may be configured to provide a semiconductor processing chemical from storage drum **102b** to chemical test POU manifold **112**

through a flow path, independent from the flow path of chemical supply line **202**, that includes one or more valves **106**, pump **108c**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like.

[0041] As shown in FIG. 2D, an example independent production and testing configuration may include chemical supply line **202** being configured to provide a semiconductor processing chemical from day tank **104** to one or more production POU manifolds **110** through one or more valves **106**, pump **108a**, a pressure tank **116**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like. Moreover, in the example independent production and testing configuration illustrated in FIG. 2D, chemical supply line **208** may be configured to provide a semiconductor processing chemical from storage drum **102a** to chemical test POU manifold **112** through a flow path, independent from the flow path of chemical supply line **202**, that includes

[0042] As shown in FIG. 2E, an example independent production and testing configuration may include chemical supply line **202** being configured to provide a semiconductor processing chemical from day tank **104** to one or more production POU manifolds **110** through one or more valves **106**, pump **108b**, a pressure tank **116**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like. Moreover, in the example independent production and testing configuration illustrated in FIG. 2E, chemical supply line **208** may be configured to provide a semiconductor processing chemical from storage drum **102b** to chemical test POU manifold **112** through a flow path, independent from the flow path of chemical supply line **202**, that includes one or more valves **106**, pump **108c**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like.

[0043] As shown in FIG. 2F, an example independent production and testing configuration may include chemical supply line **202** being configured to provide a semiconductor processing chemical from day tank **104** to one or more production POU manifolds **110** through one or more valves **106**, pump **108b**, a pressure tank **116**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like. Moreover, in the example independent production and testing configuration illustrated in FIG. 2F, chemical supply line **208** may be configured to provide a semiconductor processing chemical from storage drum **102a** to chemical test POU manifold **112** through a flow path, independent from the flow path of chemical supply line **202**, that includes one or more valves **106**, pump **108c**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like.

[0044] As shown in FIG. 2G, an example independent production and testing configuration may include chemical supply line **202** being configured to provide a semiconductor processing chemical from day tank **104** to one or more production POU manifolds **110** through one or more valves **106**, pump **108a**, a pressure tank **116**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like. Moreover, in the example independent production and testing configuration illustrated in FIG. 2G, chemical supply line **208** may be configured to provide a semiconductor processing chemical from storage drum **102b** to chemical test POU manifold **112** through a flow path, independent from the flow path of chemical supply line **202**, that includes

[0045] As shown in FIG. 2H, an example independent production and testing configuration may include chemical supply line **202** being configured to provide a semiconductor processing chemical from day tank **104** to one or more production POU manifolds **110** through one or more valves **106**, pump **108a**, a pressure tank **116**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like. Moreover, in the example independent production and testing configuration illustrated in FIG. 2H, chemical supply line **208** may be configured to provide a semiconductor processing chemical from storage drum **102a** to chemical test POU manifold **112** through a flow path, independent from the flow path of chemical supply line **202**, that includes one or more valves **106**, pump **108c**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like.

[0046] As shown in FIG. 2I, an example independent production and testing configuration may

include chemical supply line **202** being configured to provide a semiconductor processing chemical from day tank **104** to one or more production POU manifolds **110** through one or more valves **106**, pump **108b**, a pressure tank **116**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like. Moreover, in the example independent production and testing configuration illustrated in FIG. 2I, chemical supply line **208** may be configured to provide a semiconductor processing chemical from storage drum **102b** to chemical test POU manifold **112** through a flow path, independent from the flow path of chemical supply line **202**, that includes one or more valves **106**, pump **108c**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like.

[0047] As shown in FIG. 2J, an example independent production and testing configuration may include chemical supply line **202** being configured to provide a semiconductor processing chemical from day tank **104** to one or more production POU manifolds **110** through one or more valves **106**, pump **108b**, a pressure tank **116**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like. Moreover, in the example independent production and testing configuration illustrated in FIG. 2J, chemical supply line **208** may be configured to provide a semiconductor processing chemical from storage drum **102a** to chemical test POU manifold **112** through a flow path, independent from the flow path of chemical supply line **202**, that includes

[0048] As shown in FIG. 2K, an example independent production and testing configuration may include chemical supply line **202** being configured to provide a semiconductor processing chemical from day tank **104** to chemical test POU manifold **112** through one or more valves **106**, pump **108a**, a pressure tank **116**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like. Moreover, in the example independent production and testing configuration illustrated in FIG. 2K, chemical supply line **208** may be configured to provide a semiconductor processing chemical from storage drum **102b** to one or more production POU manifolds **110** through a flow path, independent from the flow path of chemical supply line **202**, that includes one or more valves **106**, pump **108c**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like.

[0049] As shown in FIG. 2L, an example independent production and testing configuration may include chemical supply line **202** being configured to provide a semiconductor processing chemical from day tank **104** to chemical test POU manifold **112** through one or more valves **106**, pump **108a**, a pressure tank **116**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like. Moreover, in the example independent production and testing configuration illustrated in FIG. 2L, chemical supply line **208** may be configured to provide a semiconductor processing chemical from storage drum **102a** to one or more production POU manifolds **110** through a flow path, independent from the flow path of chemical supply line **202**, that includes one or more valves **106**, pump **108c**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like.

[0050] As shown in FIG. 2M, an example independent production and testing configuration may include chemical supply line **202** being configured to provide a semiconductor processing chemical from day tank **104** to chemical test POU manifold **112** through one or more valves **106**, pump **108b**, a pressure tank **116**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like. Moreover, in the example independent production and testing configuration illustrated in FIG. 2M, chemical supply line **208** may be configured to provide a semiconductor processing chemical from storage drum **102b** to one or more production POU manifolds **110** through a flow path, independent from the flow path of chemical supply line **202**, that includes one or more valves **106**, pump **108c**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like.

[0051] As shown in FIG. 2N, an example independent production and testing configuration may include chemical supply line **202** being configured to provide a semiconductor processing chemical from day tank **104** to chemical test POU manifold **112** through one or more valves **106**, pump

108b, a pressure tank **116**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like. Moreover, in the example independent production and testing configuration illustrated in FIG. 2N, chemical supply line **208** may be configured to provide a semiconductor processing chemical from storage drum **102a** to one or more production POU manifolds **110** through a flow path, independent from the flow path of chemical supply line **202**, that includes one or more valves **106**, pump **108c**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like.

[0052] As shown in FIG. 2O, an example independent production and testing configuration may include chemical supply line **202** being configured to provide a semiconductor processing chemical from day tank **104** to chemical test POU manifold **112** through one or more valves **106**, pump **108a**, a pressure tank **116**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like. Moreover, in the example independent production and testing configuration illustrated in FIG. 2O, chemical supply line **208** may be configured to provide a semiconductor processing chemical from storage drum **102b** to one or more production POU manifolds **110** through a flow path, independent from the flow path of chemical supply line **202**, that includes one or more valves **106**, pump **108c**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like.

[0053] As shown in FIG. 2P, an example independent production and testing configuration may include chemical supply line **202** being configured to provide a semiconductor processing chemical from day tank **104** to chemical test POU manifold **112** through one or more valves **106**, pump **108a**, a pressure tank **116**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like. Moreover, in the example independent production and testing configuration illustrated in FIG. 2P, chemical supply line **208** may be configured to provide a semiconductor processing chemical from storage drum **102a** to one or more production POU manifolds **110** through a flow path, independent from the flow path of chemical supply line **202**, that includes one or more valves **106**, pump **108c**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like.

[0054] As shown in FIG. 2Q, an example independent production and testing configuration may include chemical supply line **202** being configured to provide a semiconductor processing chemical from day tank **104** to chemical test POU manifold **112** through one or more valves **106**, pump **108b**, a pressure tank **116**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like. Moreover, in the example independent production and testing configuration illustrated in FIG. 2Q, chemical supply line **208** may be configured to provide a semiconductor processing chemical from storage drum **102b** to one or more production POU manifolds **110** through a flow path, independent from the flow path of chemical supply line **202**, that includes one or more valves **106**, pump **108c**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like.

[0055] As shown in FIG. 2R, an example independent production and testing configuration may include chemical supply line **202** being configured to provide a semiconductor processing chemical from day tank **104** to chemical test POU manifold **112** through one or more valves **106**, pump **108b**, a pressure tank **116**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like. Moreover, in the example independent production and testing configuration illustrated in FIG. 2R, chemical supply line **208** may be configured to provide a semiconductor processing chemical from storage drum **102a** to one or more production POU manifolds **110** through a flow path, independent from the flow path of chemical supply line **202**, that includes one or more valves **106**, pump **108c**, a filter **114**, one or more pipes, fittings, and/or other plumbing fixtures, and/or the like.

[0056] As indicated above, FIGS. 2A-2R are provided as examples. Other examples may differ from what is described with regard to FIGS. 2A-2R.

[0057] FIGS. 3A and 3B are diagrams of one or more example implementations **300** described

herein. As shown in FIGS. 3A and 3B, example implementation(s) **300** may include a chemical dispensing system, such as chemical dispensing system **100** illustrated and described above in connection with FIGS. 1A and 1B.

[0058] As shown in FIG. 3A, and by reference number **302**, controller **118** may determine to filter a semiconductor processing chemical in a storage drum **102**. In some implementations, controller **118** may determine to filter the semiconductor processing chemical based on determining that a new batch of semiconductor processing chemical has been added to the storage drum **102**, based on determining that the storage drum **102** is in a standby mode (e.g., a mode in which the storage drum **102** is not supplying semiconductor processing chemical to a production POU manifold **110** or a chemical test POU manifold **112**), and/or the like. In some implementations, controller **118** may determine to filter the semiconductor processing chemical based on receiving an instruction to filter the semiconductor processing chemical in the storage drum **102**. In some implementations, the instruction may include an input from a user via an input device such as a mouse, a keyboard, a touch screen, a microphone to capture a voice command, and/or the like. In some implementations, the instruction may include an instruction received as input from another component included in chemical dispensing system **100** (e.g., another controller **118** and/or the like). In some implementations, the instruction may include an instruction received as input from a component that is remote from chemical dispensing system **100**, such as a remote computer, a cloud computing environment, and/or the like.

[0059] As further shown in FIG. 3A, and by reference number **304**, controller **118** may identify a configured quantity of filtration cycles for filtering the semiconductor processing chemical in the storage drum **102**. The configured quantity of filtration cycles may be a set and/or configured quantity of times that new batches of semiconductor processing chemical are filtered through a filtration loop (e.g., 3 times through the filtration loop, 5 times through the filtration loop, and/or the like) after being added to the storage drum **102** and prior to being used to supply semiconductor processing chemical to day tank **104**. The configured quantity of filtration cycles is used to filter each new batch of semiconductor processing chemical the same quantity of filtration cycles, thereby ensuring a similar level of filtration and a similar level of quality control across batches of semiconductor processing chemical.

[0060] Controller **118** may identify the configured quantity of filtration cycles based on information identifying the configured quantity of filtration cycles. In some implementations, the information identifying the configured quantity of filtration cycles is included in a data structure of controller **118**, such as a memory component, a storage component, an electronic database, an electronic table, an electronic file or file system, and/or the like. In some implementations, the information identifying the configured quantity of filtration cycles is included in a data structure of another device or system, such as a server, a cloud environment, and/or the like.

[0061] As further shown in FIG. 3A, and by reference number **306**, controller **118** may cause a pump **108** to circulate the semiconductor processing chemical through a filtration loop that includes one or more filters **114**. Controller **118** may cause the pump **108** to circulate the semiconductor processing chemical through the filtration loop for the configured quantity of filtration cycles. In some implementations, controller **118** may determine that a filtration cycle is complete based on the storage capacity of the storage drum **102**, the volume of semiconductor processing chemical stored in the storage drum **102**, the filtration rate of the filtration loop, and/or the like. For example, controller **118** may determine that the storage drum **102** stores 200 liters of semiconductor processing chemical and that the filtration loop is capable of processing 20 liters per minute. Accordingly, controller **118** may determine that each filtration cycle is 10 minutes.

[0062] FIG. 3B illustrates an example implementation of a filtration loop **308** for filtering a semiconductor processing chemical directly from a storage drum **102**. As shown in FIG. 3B, filtration loop **308** may include one or more valves **106**, pump **108c**, one or more pipes, one or more fittings, and/or one or more other types of plumbing fixtures, and/or the like that are

configured to circulate the semiconductor processing chemical in the storage drum **102** through a filter **114** and back into the storage drum **102** (e.g., storage drum **102b**).

[0063] In some implementations, chemical dispensing system **100** is capable of independently supplying semiconductor processing chemical in day tank **104** simultaneously with filtering the semiconductor processing chemical in the storage drum **102** through filtration loop **308**. For example, controller **118** may configure independent chemical supply lines in chemical dispensing system **100** such that chemical supply line **202** provides the semiconductor processing chemical from day tank **104** to one or more production POU manifolds **110** for production use or to chemical test POU manifold **112** for test in parallel with chemical supply line **208** cycling the semiconductor processing chemical in the storage drum **102** through filtration loop **308**. In these cases, pump **108a** or pump **108b** may pump the semiconductor processing chemical from day tank **104** through a configured set of one or more pipes, one or more fittings, and/or one or more other types of plumbing fixtures that is independent from the set of one or more pipes, one or more fittings, and/or one or more other types of plumbing fixtures through which pump **108c** pumps the semiconductor processing chemical in the storage drum **102** through filtration loop **308**.

[0064] As indicated above, FIGS. **3A** and **3B** are provided as one or more examples. Other examples may differ from what is described with regard to FIGS. **3A** and **3B**.

[0065] FIG. **4** is a diagram of example components of a device **400**. Device **400** may correspond to controller **118**. In some implementations, controller **118** may include one or more devices **400** and/or one or more components of device **400**. As shown in FIG. **4**, device **400** may include a bus **410**, a processor **420**, a memory **430**, a storage component **440**, an input component **450**, an output component **460**, and a communication interface **470**.

[0066] Bus **410** includes a component that permits communication among multiple components of device **400**. Processor **420** is implemented in hardware, firmware, and/or a combination of hardware and software. Processor **420** is a central processing unit (CPU), a graphics processing unit (GPU), an accelerated processing unit (APU), a microprocessor, a microcontroller, a digital signal processor (DSP), a field-programmable gate array (FPGA), an application-specific integrated circuit (ASIC), or another type of processing component. In some implementations, processor **420** includes one or more processors capable of being programmed to perform a function. Memory **430** includes a random access memory (RAM), a read only memory (ROM), and/or another type of dynamic or static storage device (e.g., a flash memory, a magnetic memory, and/or an optical memory) that stores information and/or instructions for use by processor **420**.

[0067] Storage component **440** stores information and/or software related to the operation and use of device **400**. For example, storage component **440** may include a hard disk (e.g., a magnetic disk, an optical disk, and/or a magneto-optic disk), a solid state drive (SSD), a compact disc (CD), a digital versatile disc (DVD), a floppy disk, a cartridge, a magnetic tape, and/or another type of non-transitory computer-readable medium, along with a corresponding drive.

[0068] Input component **450** includes a component that permits device **400** to receive information, such as via user input (e.g., a touch screen display, a keyboard, a keypad, a mouse, a button, a switch, and/or a microphone). Additionally, or alternatively, input component **450** may include a component for determining location (e.g., a global positioning system (GPS) component) and/or a sensor (e.g., an accelerometer, a gyroscope, an actuator, another type of positional or environmental sensor, and/or the like). Output component **460** includes a component that provides output information from device **400** (via, e.g., a display, a speaker, a haptic feedback component, an audio or visual indicator, and/or the like).

[0069] Communication interface **470** includes a transceiver-like component (e.g., a transceiver, a separate receiver, a separate transmitter, and/or the like) that enables device **400** to communicate with other devices, such as via a wired connection, a wireless connection, or a combination of wired and wireless connections. Communication interface **470** may permit device **400** to receive information from another device and/or provide information to another device. For example,

communication interface **470** may include an Ethernet interface, an optical interface, a coaxial interface, an infrared interface, a radio frequency (RF) interface, a universal serial bus (USB) interface, a Wi-Fi interface, a cellular network interface, and/or the like.

[0070] Device **400** may perform one or more processes described herein. Device **400** may perform these processes based on processor **420** executing software instructions stored by a non-transitory computer-readable medium, such as memory **430** and/or storage component **440**. As used herein, the term “computer-readable medium” refers to a non-transitory memory device. A memory device includes memory space within a single physical storage device or memory space spread across multiple physical storage devices.

[0071] Software instructions may be read into memory **430** and/or storage component **440** from another computer-readable medium or from another device via communication interface **470**. When executed, software instructions stored in memory **430** and/or storage component **440** may cause processor **420** to perform one or more processes described herein. Additionally, or alternatively, hardware circuitry may be used in place of or in combination with software instructions to perform one or more processes described herein. Thus, implementations described herein are not limited to any specific combination of hardware circuitry and software.

[0072] The number and arrangement of components shown in FIG. **4** are provided as an example. In practice, device **400** may include additional components, fewer components, different components, or differently arranged components than those shown in FIG. **4**. Additionally, or alternatively, a set of components (e.g., one or more components) of device **400** may perform one or more functions described as being performed by another set of components of device **400**.

[0073] FIG. **5** is a flow chart of an example process **500** for dispensing a semiconductor processing chemical. In some implementations, one or more process blocks of FIG. **5** may be performed by a controller (e.g., controller **180**, device **400**, and/or the like). In some implementations, one or more process blocks of FIG. **5** may be performed by another device or a group of devices separate from or including the controller, such as another component of a chemical dispensing system (e.g., chemical dispensing system **100**) and/or the like.

[0074] As shown in FIG. **5**, process **500** may include determining that a test is to be performed for a semiconductor processing chemical in a storage drum of a chemical dispensing system (block **510**). For example, the controller (e.g., using processor **420**, memory **430**, storage component **440**, input component **450**, output component **460**, communication interface **470**, and/or the like) may determine that a test is to be performed for a semiconductor processing chemical in a storage drum (e.g., storage drum **102**) of a chemical dispensing system (e.g., chemical dispensing system **100**), as described above.

[0075] As further shown in FIG. **5**, process **500** may include causing, based on determining that the test is to be performed, the chemical dispensing system to be transitioned from a production supply configuration to an independent production and test configuration, in which a first chemical supply line of the chemical dispensing system is configured to provide a semiconductor processing chemical in a day tank to a POU or a chemical test POU, and a second chemical supply line of the chemical dispensing system is configured to provide the semiconductor processing chemical in the storage drum to the chemical test POU or the production POU based on whether the first chemical supply line is configured to provide the semiconductor processing chemical in the day tank to the production POU or the chemical test POU (block **520**). For example, the controller (e.g., using processor **420**, memory **430**, storage component **440**, input component **450**, output component **460**, communication interface **470**, and/or the like) may cause, based on determining that the test is to be performed, the chemical dispensing system to be transitioned from a production supply configuration to an independent production and test configuration, as described above. In some implementations, in the independent production and test configuration, a first chemical supply line (e.g., chemical supply line **202**) of the chemical dispensing system is configured to provide a semiconductor processing chemical in a day tank (e.g., day tank **104**) to a production POU (e.g.,

via a production POU manifold **110**) or a chemical test POU (e.g., via a chemical test POU manifold **112**). In some implementations, in the independent production and test configuration, a second chemical supply line (e.g., chemical supply line **208**) of the chemical dispensing system is configured to provide the semiconductor processing chemical in the storage drum to the chemical test POU or the production POU based on whether the first chemical supply line is configured to provide the semiconductor processing chemical in the day tank to the production POU or the chemical test POU.

[0076] Process **500** may include additional implementations, such as any single implementation or any combination of implementations described below and/or in connection with one or more other processes described elsewhere herein.

[0077] In a first implementation, in the independent production and test configuration, the first chemical supply line is configured to provide the semiconductor processing chemical in the day tank to the production POU, and the second chemical supply line is configured to provide the semiconductor processing chemical in the storage drum to the chemical test POU, and the process **500** includes causing a first pump (e.g., pump **108a**, pump **108b**, and/or the like) to provide the semiconductor processing chemical in the day tank to the production POU through the first chemical supply line, and causing a second pump (e.g., pump **108c**) to provide the semiconductor processing chemical in the storage drum to the chemical test POU through the second chemical supply line. In a second implementation, alone or in combination with the first implementation, determining that the test is to be performed includes determining that the test is to be performed based on determining that a new batch of semiconductor processing chemical was added to the storage drum.

[0078] In a third implementation, alone or in combination with one or more of the first and second implementations, determining that the test is to be performed includes determining that the test is to be performed based on receiving an instruction to perform the test. In a fourth implementation, alone or in combination with one or more of the first through third implementations, in the independent production and test configuration, the first chemical supply line is configured to provide the semiconductor processing chemical in the day tank to the chemical test POU, and the second chemical supply line is configured to provide the semiconductor processing chemical in the storage drum to the production POU, and process **500** includes causing a first pump (e.g., pump **108a**, pump **108b**, and/or the like) to provide the semiconductor processing chemical in the day tank to the chemical test POU through the first chemical supply line, and causing a second pump (e.g., pump **108c**) to provide the semiconductor processing chemical in the storage drum to the production POU through the second chemical supply line.

[0079] In a fifth implementation, alone or in combination with one or more of the first through fourth implementations, process **500** includes determining to filter the semiconductor processing chemical in the storage drum, identifying a configured quantity of filtration cycles for filtering the semiconductor processing chemical in the storage drum, and causing a pump to circulate the semiconductor processing chemical in the storage drum through a filter (e.g., filter **114**) included in the chemical dispensing system for a quantity of filtration cycles equal to the configured quantity of filtration cycles. In a sixth implementation, alone or in combination with one or more of the first through fifth implementations, process **500** includes determining, based on a test of the semiconductor processing chemical in the storage drum, one or more parameters for the semiconductor processing chemical in the storage drum, determining that the one or more parameters satisfy one or more test thresholds, and causing, based on determining that the one or more parameters for the semiconductor processing chemical in the storage drum satisfies the one or more test thresholds, the chemical dispensing system to be transitioned from the independent production and test configuration to the production supply configuration, in which the first chemical supply line is connected to the day tank, the production POU, and the chemical test POU.

[0080] In a seventh implementation, alone or in combination with one or more of the first through

sixth implementations, process **500** includes determining, based a first test of the semiconductor processing chemical in the storage drum, one or more parameters for the semiconductor processing chemical in the storage drum, determining that the one or more parameters do not satisfy one or more test thresholds, causing the second supply line to be flushed based on determining that the one or more parameters for the semiconductor processing chemical in the storage drum does not satisfy the one or more test thresholds, and causing a second test to be performed for the semiconductor processing chemical in the storage drum after the second chemical supply line is flushed.

[0081] Although FIG. 5 shows example blocks of process **500**, in some implementations, process **500** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. 5. Additionally, or alternatively, two or more of the blocks of process **500** may be performed in parallel.

[0082] FIG. 6 is a flow chart of an example process **600** for filtering a semiconductor processing chemical. In some implementations, one or more process blocks of FIG. 6 may be performed by a controller (e.g., controller **180**, device **400**, and/or the like). In some implementations, one or more process blocks of FIG. 6 may be performed by another device or a group of devices separate from or including the controller, such as another component of a chemical dispensing system (e.g., chemical dispensing system **100**) and/or the like.

[0083] As shown in FIG. 6, process **600** may include determining to filter a semiconductor processing chemical in a storage drum included in a chemical dispensing system (block **610**). For example, the controller (e.g., using processor **420**, memory **430**, storage component **440**, input component **450**, output component **460**, communication interface **470**, and/or the like) may determine to filter a semiconductor processing chemical in a storage drum (e.g., storage drum **102**) included in a chemical dispensing system (e.g., chemical dispensing system **100**), as described above.

[0084] As further shown in FIG. 6, process **600** may include identifying a configured quantity of filtration cycles for filtering the semiconductor processing chemical (block **620**). For example, the controller (e.g., using processor **420**, memory **430**, storage component **440**, input component **450**, output component **460**, communication interface **470**, and/or the like) may identify a configured quantity of filtration cycles for filtering the semiconductor processing chemical, as described above.

[0085] As further shown in FIG. 6, process **600** may include causing a pump to circulate the semiconductor processing chemical through a filter included in the chemical dispensing system for a quantity of filtration cycles equal to the configured quantity of filtration cycles (block **630**). For example, the controller (e.g., using processor **420**, memory **430**, storage component **440**, input component **450**, output component **460**, communication interface **470**, and/or the like) may cause a pump (e.g., pump **108c**) to circulate the semiconductor processing chemical through a filter (e.g., filter **114**) included in the chemical dispensing system for a quantity of filtration cycles equal to the configured quantity of filtration cycles, as described above.

[0086] Process **600** may include additional implementations, such as any single implementation or any combination of implementations described below and/or in connection with one or more other processes described elsewhere herein.

[0087] In a first implementation, process **600** includes causing the pump to circulate the semiconductor processing chemical through a first supply line (e.g., chemical supply line **208**) that is independent from a second supply line (e.g., chemical supply line **202**) that is used to provide a semiconductor processing chemical in a day tank (e.g., day tank **104**) to a production POU (e.g., via a production POU manifold **110**). In a second implementation, alone or in combination with the first implementation, process **600** includes causing another pump (e.g., pump **108a**, pump **108b**, and/or the like) to provide the semiconductor processing chemical in the day tank to the production POU through the second supply line. In a third implementation, alone or in combination with one

or more of the first and second implementations, process **600** includes determining that the storage drum is in a standby mode; and determining to filter the semiconductor processing chemical based on determining that the storage drum is in the standby mode.

[0088] Although FIG. **6** shows example blocks of process **600**, in some implementations, process **600** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **6**. Additionally, or alternatively, two or more of the blocks of process **600** may be performed in parallel.

[0089] FIG. **7** is a flowchart of an example process **700** for dispensing a semiconductor processing chemical. In some implementations, one or more process blocks of FIG. **7** may be performed by a controller (e.g., controller **180**, device **400**, and/or the like). In some implementations, one or more process blocks of FIG. **7** may be performed by another device or a group of devices separate from or including the controller, such as another component in a chemical dispensing system (e.g., chemical dispensing system **100**) and/or the like.

[0090] As shown in FIG. **7**, process **700** may include causing a chemical supply system to operate in a production supply configuration (block **710**). For example, the controller (e.g., using processor **420**, memory **430**, storage component **440**, input component **450**, output component **460**, communication interface **470**, and/or the like) may cause a chemical dispensing system (e.g., chemical dispensing system **100**) to operate in a production supply configuration, as described above. In some implementations, in the production supply configuration, a first chemical supply line (e.g., chemical supply line **202**) is connected to a day tank (e.g., day tank **104**) of the chemical dispensing system, one or more production POU manifolds (e.g., production POU manifold(s) **110**), and a chemical test POU manifold (e.g., chemical test POU manifold **112**).

[0091] As further shown in FIG. **7**, process **700** may include determining to perform a test based on a new semiconductor processing chemical batch being added to a storage drum of the chemical dispensing system, based on a test-run request, based on a maintenance component test for a component added to or replaced in the chemical dispensing system, and/or the like (block **720**). For example, the controller (e.g., using processor **420**, memory **430**, storage component **440**, input component **450**, output component **460**, communication interface **470**, and/or the like) may determine to perform a test based on a new semiconductor processing chemical batch being added to a storage drum (e.g., storage drum **102**) of the chemical dispensing system, based on a test-run request, based on a maintenance component test for a component (e.g., a replacement pump **108**, a replacement valve **106**, and/or the like) added to or replaced in the chemical dispensing system, and/or the like, as described above. In some implementations, the test may include supplying a semiconductor processing chemical to the chemical test POU manifold so that wafer test data may be generated based on the semiconductor processing chemical.

[0092] As further shown in FIG. **7**, process **700** may include causing the chemical supply system to operate in an independent production and test configuration (block **730**). For example, the controller (e.g., using processor **420**, memory **430**, storage component **440**, input component **450**, output component **460**, communication interface **470**, and/or the like) may cause the chemical supply system to operate in an independent production and test configuration, as described above. In some implementations, in the independent production and test configuration, the first chemical supply line (e.g., chemical supply line **202**) of the chemical dispensing system is configured to provide the semiconductor processing chemical in the day tank to the production POU manifold or the chemical test POU manifold, and a second chemical supply line (e.g., chemical supply line **208**) of the chemical dispensing system is configured to provide a semiconductor processing chemical in a storage drum (e.g., storage drum **102**) of the chemical dispensing system to the chemical test POU manifold or the production POU manifold based on whether the first chemical supply line is configured to provide the semiconductor processing chemical in the day tank to the production POU manifold or the chemical test POU manifold.

[0093] As further shown in FIG. **7**, process **700** may include generating wafer test data (block **740**).

For example, the controller (e.g., using processor **420**, memory **430**, storage component **440**, input component **450**, output component **460**, communication interface **470**, and/or the like) may generate wafer test data, as described above. In some implementations, the wafer test data may be generated based on performing the test while the chemical dispensing system is operating in the independent production and test configuration. In some implementations, the controller may determine whether the test passed or failed based on the wafer test data.

[0094] As further shown in FIG. 7, process **700** may include automatically performing a flush of the chemical supply line that was used to supply the semiconductor processing chemical to the chemical test POU manifold (block **750**). For example, the controller (e.g., using processor **420**, memory **430**, storage component **440**, input component **450**, output component **460**, communication interface **470**, and/or the like) may automatically perform a flush of the chemical supply line that was used to supply the semiconductor processing chemical to the chemical test POU manifold, as described above. In some implementations, the flush of the chemical supply line that was used to supply the semiconductor processing chemical to the chemical test POU manifold may be performed based on a determination that the test failed (block **740**—Fail).

[0095] As further shown in FIG. 7, process **700** may include automatically restarting the test (block **760**). For example, the controller (e.g., using processor **420**, memory **430**, storage component **440**, input component **450**, output component **460**, communication interface **470**, and/or the like) may automatically restart the test, as described above. In some implementations, the actions associated with blocks **740-760** may be performed until the controller determines that the test passes. In some implementations, the controller may cause the semiconductor processing chemical to be filtered for one or more filtration cycles based on determining that the test failed.

[0096] As further shown in FIG. 7, process **700** may include automatically causing the chemical supply system to transition to the production supply configuration (block **770**). For example, the controller (e.g., using processor **420**, memory **430**, storage component **440**, input component **450**, output component **460**, communication interface **470**, and/or the like) may automatically cause the chemical supply system to transition to the production supply configuration, as described above. In some implementations, the controller may automatically cause the chemical supply system to transition to the production supply configuration based on determining that the test passed (block **740**—Pass).

[0097] Process **700** may include additional implementations, such as any single implementation or any combination of implementations described in connection with one or more other processes described elsewhere herein. Although FIG. 7 shows example blocks of process **700**, in some implementations, process **700** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. 7. Additionally, or alternatively, two or more of the blocks of process **700** may be performed in parallel.

[0098] In this way, chemical dispensing system **100** is capable of simultaneously supplying a semiconductor processing chemical for production and testing through the use of independent chemical supply lines **202** and **208**, which reduces production downtime of an associated semiconductor process, increases throughput and capability of the semiconductor process, and/or the like. Moreover, the capability to simultaneously supply the semiconductor processing chemical for production and testing allows for an increased quantity of semiconductor processing chemical batches to be tested with minimal impact to production, which increases quality control over the semiconductor processing chemical. In addition, the independent chemical supply lines **202** and **208** may be used to supply the semiconductor processing chemical to production while independently filtering semiconductor processing chemical directly from a storage drum **102** through a filtration loop **308**.

[0099] As described in greater detail above, some implementations described herein provide a chemical dispensing system. The chemical dispensing system includes a first chemical supply line configured to provide a semiconductor processing chemical from a day tank to one or more

production POU manifolds or a chemical test POU manifold. The chemical dispensing system includes a second chemical supply line configured to provide a semiconductor processing chemical from a storage drum to the chemical test POU manifold or the one or more production POU manifolds based on whether the first chemical supply line is configured to provide the semiconductor processing chemical from the day tank to the one or more POU manifolds or the chemical test POU manifold. A pump used for the first chemical supply line is different from a pump used for the second chemical supply line.

[0100] As described in greater detail above, some implementations described herein provide a method. The method includes determining, by one or more processors, that a test is to be performed for a semiconductor processing chemical in a storage drum of a chemical dispensing system. The method includes causing, by the one or more processors and based on determining that the test is to be performed, the chemical dispensing system to be transitioned from a production supply configuration to an independent production and test configuration. In the independent production and test configuration, a first chemical supply line of the chemical dispensing system is configured to provide a semiconductor processing chemical in a day tank to a production POU or a chemical test POU. In the independent production and test configuration, a second chemical supply line of the chemical dispensing system is configured to provide the semiconductor processing chemical in the storage drum to the chemical test POU or the production POU based on whether the first chemical supply line is configured to provide the semiconductor processing chemical in the day tank to the production POU or the chemical test POU.

[0101] As described in greater detail above, some implementations described herein provide a device. The device includes one or more memories and one or more processors, communicatively coupled to the one or more memories. The one or more processors are to determine to filter a semiconductor processing chemical in a storage drum included in a chemical dispensing system. The one or more processors are to identify a configured quantity of filtration cycles for filtering the semiconductor processing chemical. The one or more processors are to cause a pump to circulate the semiconductor processing chemical through a filter included in the chemical dispensing system for a quantity of filtration cycles equal to the configured quantity of filtration cycles.

[0102] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A chemical dispensing system, comprising: a first chemical supply line configured to: provide a semiconductor processing chemical in a day tank to a production point of use (POU) manifold when the chemical dispensing system is in a first independent production and test configuration, and ,provide the semiconductor processing chemical in the day tank to a chemical test POU manifold when the chemical dispensing system is in a second independent production and test configuration; and a second chemical supply line configured to: provide the semiconductor processing chemical in a storage drum to the chemical test POU manifold when the chemical dispensing system is in the first independent production and test configuration, and provide the semiconductor processing chemical in the storage drum to the production POU manifold when the chemical dispensing system is in the second independent production and test configuration, wherein the first chemical supply line is different from the second chemical supply line when the

chemical dispensing system is in the second independent production and test configuration, and wherein the first chemical supply line is configured to simultaneously provide the semiconductor processing chemical in the day tank to the chemical test POU manifold while the second chemical supply line provides the semiconductor processing chemical in the storage drum to the production POU manifold when the chemical dispensing system is in the second independent production and test configuration.

2. The chemical dispensing system of claim 1, further comprising: a controller configured to cause the chemical dispensing system to be transitioned from a production supply configuration to the second independent production and test configuration.

3. The chemical dispensing system of claim 2, wherein the controller, to cause the chemical dispensing system to be transitioned from the production supply configuration to the second independent production and test configuration, is configured to one or more of: close one or more first valves, open one or more second valves, activate one or more first pumps, deactivate one or more second pumps, reconfigure one or more filters, rearrange the one or more filters, reconfigure one or more pipes, rearrange the one or more pipes, reconfigure one or more fittings, or rearrange the one or more fittings.

4. The chemical dispensing system of claim 1, wherein the first chemical supply line is connected to the day tank, the production POU manifold, and the chemical test POU manifold when the chemical dispensing system is in a production supply configuration.

5. The chemical dispensing system of claim 1, further comprising: a controller configured to generate wafer test data based on a test performed while the chemical dispensing system is the first independent production and test configuration.

6. The chemical dispensing system of claim 5, wherein the controller is further configured to cause the chemical dispensing system to transition to a production supply configuration based on a result of the test.

7. The chemical dispensing system of claim 1, wherein the first chemical supply line is configured to use one or more of a different valve, a different pump, a different filter, or a different pipe from the second chemical supply line when the chemical dispensing system is in the second independent production and test configuration.

8. A method, comprising: determining, by one or more processors, that a test is to be performed for a semiconductor processing chemical in a storage drum of a chemical dispensing system; causing, by the one or more processors and based on determining that the test is to be performed, the chemical dispensing system to operate in an independent production and test configuration; causing, by the one or more processors and based on the chemical dispensing system operating in the independent production and test configuration, a first chemical supply line of the chemical dispensing system to provide the semiconductor processing chemical in a day tank to a first one of a production point of use (POU) manifold or a chemical test POU manifold; and causing, by the one or more processors and based on the chemical dispensing system operating in the independent production and test configuration, a second chemical supply line of the chemical dispensing system to provide the semiconductor processing chemical in the storage drum to a second one of the production POU manifold or the chemical test POU manifold.

9. The method of claim 8, further comprising: causing, based on the chemical dispensing system being configured in the independent production and test configuration, the first chemical supply line to provide the semiconductor processing chemical in the day tank to the production POU manifold simultaneously with the second chemical supply line providing the semiconductor processing chemical in the storage drum to the chemical test POU manifold.

10. The method of claim 8, further comprising: causing, based on the chemical dispensing system being configured in the independent production and test configuration, the first chemical supply line to provide the semiconductor processing chemical in the day tank to the chemical test POU manifold simultaneously with the second chemical supply line providing the semiconductor

processing chemical in the storage drum to the production POU manifold.

11. The method of claim 8, further comprising: generating wafer test data based on performing a test while the chemical dispensing system is operating in the independent production and test configuration.

12. The method of claim 11, further comprising: performing a flushing of the first chemical supply line or the second chemical supply line based on the wafer test data indicating that the test failed.

13. The method of claim 12, further comprising: restarting the test based on performing the flushing.

14. The method of claim 11, further comprising: causing the chemical dispensing system to transition to a production supply configuration based on the wafer test data indicating that the test passed.

15. A device, comprising: one or more memories; and one or more processors, coupled to the one or more memories, configured to: determine that a test is to be performed for a semiconductor processing chemical in a storage drum of a chemical dispensing system; cause, based on a determination that the test is to be performed, the chemical dispensing system to operate in an independent production and test configuration; cause, based on the independent production and test configuration, a first chemical supply line of the chemical dispensing system to provide the semiconductor processing chemical in a day tank to a first one of a production point of use (POU) manifold or a chemical test POU manifold; and cause, based on the independent production and test configuration, a second chemical supply line of the chemical dispensing system to provide the semiconductor processing chemical in the storage drum to a second one of the production POU manifold or the chemical test POU manifold.

16. The device of claim 15, wherein the one or more processors are further configured to: cause, based on the independent production and test configuration, the first chemical supply line to provide the semiconductor processing chemical in the day tank to the production POU manifold simultaneously with the second chemical supply line providing the semiconductor processing chemical in the storage drum to the chemical test POU manifold.

17. The device of claim 15, wherein the one or more processors are further configured to: cause, based on the independent production and test configuration, the first chemical supply line to provide the semiconductor processing chemical in the day tank to the chemical test POU manifold simultaneously with the second chemical supply line providing the semiconductor processing chemical in the storage drum to the production POU manifold.

18. The device of claim 15, wherein the one or more processors are further configured to: generate wafer test data based on a test of the chemical dispensing system in the independent production and test configuration.

19. The device of claim 18, wherein the one or more processors are further configured to: perform a flushing of the first chemical supply line or the second chemical supply line based on the wafer test data indicating that the test failed.

20. The device of claim 18, wherein the one or more processors are further configured to: cause the chemical dispensing system to transition to a production supply configuration based on the wafer test data indicating that the test passed.
